Counting

6

Introduction

- Combinatorics: the study of arrangements of objects, is an important part of discrete mathematics.
- Counting problems: permutations and combinations
- Counting problems arise throughout mathematics and computer science.

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6.1 The Basics of Counting

- Basic counting principles
  - The sum rule
  - The product rule
- The inclusion-exclusion principle for two sets （Subtraction rule)
- The division rule
- Tree diagram

The sum rule

- Suppose there are two tasks:
  - The first task can be done in n_1 ways
  - The second task can be done in n_2 ways
  - These tasks cannot be done at the same time
- Then there are $(n_1 + n_2)$ ways to do either task.

- Extend the sum rule to more than two tasks:

$$n_1 + n_2 + \ldots + n_m$$

Phrased in terms of sets

- No two of the tasks can be done at the same time: sets are disjoint

$$|A_1 \cup A_2 \cup \ldots \cup A_m| = |A_1| + |A_1| + \ldots + |A_m|$$

- If $A_1, A_2, \ldots, A_m$ have common elements, then we should use the *Inclusion-Exclusion principle*.

Examples

- Suppose that you have to select a representative to a university committee
  - Either a member of the mathematics faculty or a mathematics major student can be selected.
  - There are 37 member of the mathematics faculty and 83 mathematics major students.

  How many different choices are there for selecting the representative?

- A student has to do a computer project
  - He can choose a project from one of three lists.
  - The three lists contain 23, 15, and 19 possible projects, respectively.

  How many possible projects are there to choose from?

Counting the number of elements in *A*, *A* = {length 10 bit strings with 0-streak of length exactly 5}.

Solution:

- **Since the set *A* can be break up into the following case.**
- $A1 = \{000001****\}$ (* is either 0 or 1)
- $A2 = \{1000001***\}$
- $A3 = \{*1000001**\}$
- $A4 = \{**1000001*\}$
- $A5 = \{***1000001\}$
- $A6 = \{****100000\}$
- Apply the sum rule:
- $$|A| = |A1| + |A2| + |A3| + |A4| + |A5| + |A6|$$

The product rule

- Suppose that a procedure can be broken down into two tasks:
  - The first task can be done in n_1 ways
  - The second task can be done in n_2 ways after the first task has been done
- Then there are $n_1 n_2$ ways to do the procedure.

- Extension to more than two tasks:

$$n_1 \, n_2 \, \ldots \, n_m$$

- Product rule: Cartesian product

$$|A_1 \times A_2 \times \ldots \times A_m| = |A_1| \times |A_1| \times \ldots \times |A_m|$$

Examples

- The chairs of an auditorium are to be labeled with a letter and a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

$$26 \times 100 = 2600$$

- How many different bit strings are there of length seven?

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 = 128$$

Combining Sum and Product Rules

- A programming language can have 1 or 2 character case-insensitive variable names. Each variable name must start with a letter. There are five 2-character reserved words that can not be used. How many different variable names are there?

  $26 + (26*36 - 5) = 957$

- Each user on a computer system has a password
  - Each password is six to eight characters long
  - Each character is an uppercase letter or a digit
  - Each password must contain at least one digit

  How many possible passwords are there?

  $P = P_6+P_7+P_8$

  $P_6 = 36^6 - 26^6$

Counting Internet addresses

Class A: $(2^7-1)\times(2^{24}-2)$

Class B: $2^{14}\times(2^{16}-2)$

Class C: $2^{21}\times(2^8-2)$

3,737,091,842

- ***IP address (IPv4):***
  - **A string of 32 bits**
  - **network number (netid)+ l**
  - **The IP address are classifi**

Bit number	0	1	2	3	4	8	16	24	31
Class A	0	netid: 7-bit					hostid: 24-bit		
Class B	1	0	netid: 14-bit					hostid : 16-bit	
Class C	1	1	0	netid: 21-bit					hostid : 8-bit
Class D	1	1	1	0	Multicast Address				
Class E	1	1	1	1	0	Address			

Example

- Choose three different numbers from the integers between 1 to 300 such that the sum of the three integers can be divisible by 3. How many the ways are there?

Solution:

$$A = \{x \mid 1 \le x \le 300,\ x \pmod 3 = 1\}$$

$$B = \{x \mid 1 \le x \le 300,\ x \pmod 3 = 2\}$$

$$C = \{x \mid 1 \le x \le 300,\ x \pmod 3 = 0\}$$

$$|A| = |B| = |C| = 100$$

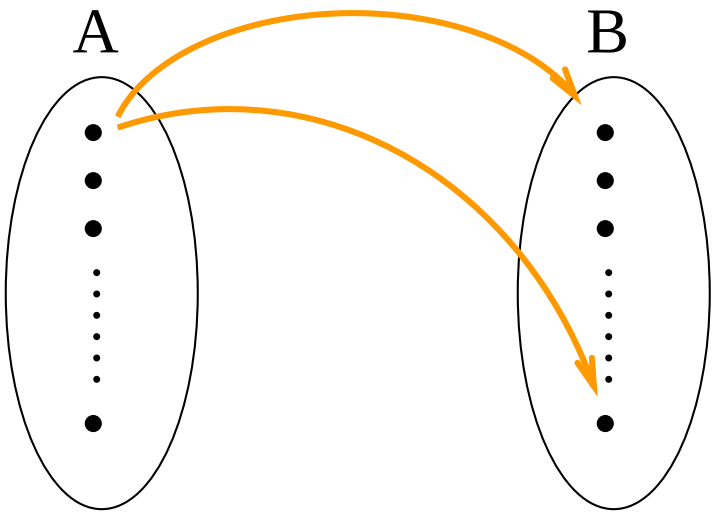
(1) All of the three numbers are chosen form the set A C_{100}^3

(2) All of the three numbers are chosen form the set B C_{100}^3

(3) All of the three numbers are chosen form the set C C_{100}^3

(4) Chose one number form the set A, B, C $C_{100}^1 \times C_{100}^1 \times C_{100}^1$

How many functions are there from a set with *m* elements to one with *n* elements?



- ***Solution:***

  By the product rule there are $n \cdot n \cdot \ldots \cdot n = n^m$ functions from a set with *m* elements to one with *n* elements

How many one-to-one functions are there from a set with *m* elements to one with *n* elements?

- ***Solution:***

(1) $m > n$

There are no one-to-one functions from a set with *m* elements to one with *n* elements.

(2) $m \leq n$

$n(n-1)(n-2)\ldots(n-m+1)$

The inclusion-exclusion principle

- Question: What happens when two tasks can be done at the same time, but we want to count the number of ways to do one of the two tasks?
- Answer: We add the number of ways to do each of the two tasks and then subtract the number of ways to do both tasks.

- How many bit strings of length eight either start with a 1 or end with the two bits 00?

  1st Task: Construct string beginning with a 1.

  2nd Task: Construct string ending with 00.

  Both task: Construct string that begins with a 1 and ends with 00.

  $2^7+2^6-2^5=160$

Division Rule

Division Rule: There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

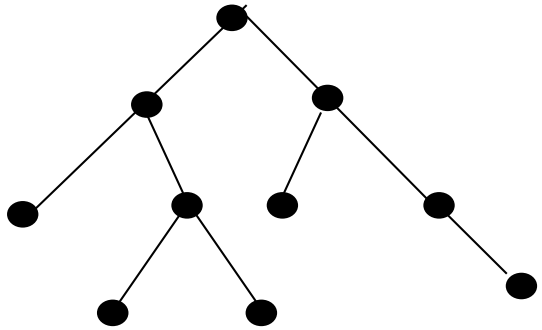
- **Restated in terms of sets: If the finite set A is the union of n pairwise disjoint subsets each with d elements, then $n = |A|/d$.**
- **In terms of functions: If f is a function from A to B, where both are finite sets, and for every value $y \in B$ there are exactly d values $x \in A$ such that $f(x) = y$, then $|B| = |A|/d$.**

Example: How many ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left and right neighbor?

Solution: Number the seats around the table from 1 to 4 proceeding clockwise. There are four ways to select the person for seat 1, 3 for seat 2, 2, for seat 3, and one way for seat 4. Thus there are 4! = 24 ways to order the four people. But since two seatings are the same when each person has the same left and right neighbor, for every choice for seat 1, we get the same seating.

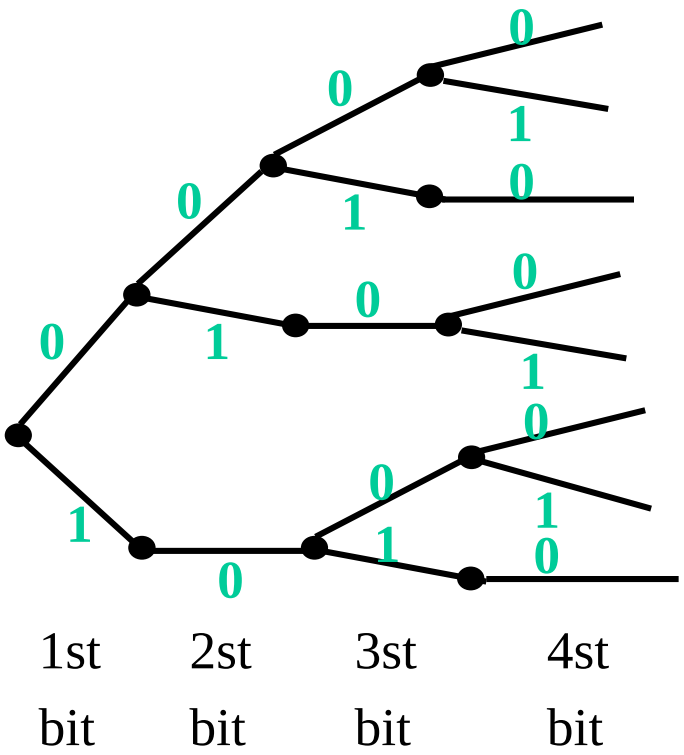
Therefore, by the division rule, there are 24/4 = 6 different seating arrangements.

Tree Diagrams



- To use trees in counting, we use a branch to represent each possible choice. We represent the possible outcomes by the leaves.

How many bit strings of length four do not have two consecutive 1s?



- ***Solution:***

There are eight bit strings of length four without two consecutive 1s.

Exercises

- **P.416: 32, 50-53**

6.2 The Pigeonhole Principle

- The pigeonhole principle and the Inclusion-Exclusion principle are two basic counting principle.
- The pigeonhole principle states that if there are more pigeons than pigeonholes, then there must be at least one pigeonhole with at least two pigeons in it.

- Theorem: If k+1 or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

Examples

- Example 1: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.
- Example 2: How many students must be in a class to guarantee that at least two students receive the same score on the final exam if the exam is graded on a scale from 0 to 100 points? Possible score: 101 Students:102
- Example 3: Among any group of 11 integers, there are two integers a and b such that $10 \mid a-b$.

The formal statement of the pigeonhole principle

If *f* is a function for *A* to *B*, where *A* and *B* are finite sets with $|A| > |B|$, then there are elements a_1, a_2 in *A*($a_1 \neq a_2$) such that $f(a_1) = f(a_2)$.

Proof:

$\forall a_1, a_2 \in A, \quad a_1 \neq a_2$,

If $f(a_1) \neq f(a_2)$, then $|A| = |f(A)| \leq |B|$.

***Corollary* 1** ***A* function *f* from a set with *k*+1 or more elements to a set with *k* elements is not one-to-one.**

The generalized pigeonhole principle

- Theorem: If *N* objects are placed into *k* boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

- Phrased in terms of functions:

$f : A \to B$, If $\left\lceil \frac{|A|}{|B|} \right\rceil = i$, then there must exist elements $a_1, a_2, \cdots, a_i \in A$ such that

$$f(a_1) = f(a_2) = \cdots = f(a_i) = b \in B$$

Examples

- Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.
- Example: What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

$$N=5\times 5+1 = 26$$

- Example: a) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen? b) How many must be selected to guarantee that at least three hearts are selected? (9,42)

Example

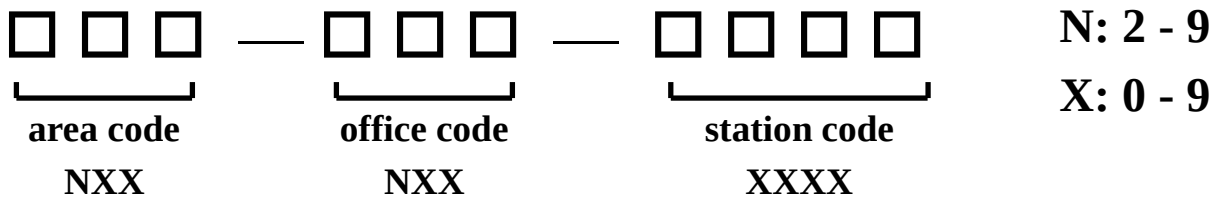
- What is the least number of area codes needed to guarantee that the 25 million phones in a state have distinct 10-digit telephone numbers?
- Solution:

  The number of phone numbers of the form NXX-XXXX

  $8 \times 10^6 = 8{,}000{,}000$

  By the generalized pigeonhole principle, among 25 million phones, at least $\lceil 25/8 \rceil = 4$ of them must have identical numbers.

  Hence, at least 4 area codes are required to.



Examples

- Example: Suppose that a computer laboratory has 15 workstations and ten servers. A cable can be used to directly connect a workstation to a server. Only one direct connection to a server can be active at any time. We want to guarantee that at any time any set of ten or fewer workstations can simultaneously access different servers via direct connections. Although we could do this by connecting every workstation directly to every server, what is the minimum number of direct connections needed to achieve this goal? (60)

Some elegant applications of the pigeonhole principle

- During 11 weeks football games will be held at least 1 game a day, but at most 12 games be arranged each week. Show that there must be a period of some number of consecutive days during which exactly 21 games must be played.

x_i : the number of football games helded on the ith day

$$a_i = \sum_{k=1}^{i} x_k \qquad 1 \le a_1 < a_2 ... < a_{77} \le 12 \times 11 = 132$$

$$c_i = a_i + 21 \qquad c_1 < c_2 < ... < c_{77} \le 132 + 21 = 153$$

$$A = \{a_1, a_2, ...a_{77}, c_1, c_2, ...c_{77}\} \qquad B = \{1,2,...,153\}$$

$$\exists i \neq j \quad such \quad that \quad a_i = c_j$$

$$a_i = a_j + 21$$

$$a_i - a_j = x_i + x_{i-1} + ... + x_{j+1} = 21$$

Example

- Show that among any n+1 positive integers not exceeding 2n there must be an integer that divides one of the other integer.

  Let n+1 positive integers be $a_1, a_2, \cdots, a_{n+1} (1 \le a_i \le 2n)$

  Write a_i(i=1,2,…,n+1) as $2^{k_i} q_i$ **, where k_i is a nonnegative integer and q_i is odd positive integers less than 2n.**

  Since there are only n odd positive integers less than 2n, by the pigeonhole principle it follows that there exist integers i and j such that q_i=q_j=q,

  then $a_i = 2^{k_i} q$ and $a_j = 2^{k_j} q$

  It follows that if $a_i < a_j$, then $a_i \mid a_j$, while if $a_j < a_i$, then $a_j \mid a_i$.

Example

- Theorem: Every sequence of n^2+1 distinct real numbers contains a subsequence of length $n+1$ that is either strictly increasing or strictly decreasing.

Proof:

let $a_1, a_2, \cdots, a_{n^2+1}$ be a sequence of n^2+1 distinct real numbers.

Associate an ordered pair with each term a_k of the sequence : (i_k, d_k)

i_k is the length of the longest increasing subsequence starting at a_k.

d_k is the length of the longest decreasing subsequence starting at a_k.

Suppose that there are no increasing or decreasing subsequences of length $n+1$.

Then i_k and d_k are both positive integers less tahn or equal to n.

Hence, by the product rule there are n^2 possible ordered pair for (i_k, d_k)

By the pigeonhole principle, two of these

Continue

By the pigeonhole principle, two of these n^2+1 ordered pairs are equal.

This is impossible, because the terms of the sequence are distinct,

either $a_s < a_t$ or $a_s > a_t$.

if $a_s < a_t$, then since $i_s = i_t$, an increasing subsequence of length $i_t + 1$

can be built starting at a_s, by taking a_s followed by an increasing subsequence of

length i_t beginning at a_t. This is a contradiction.

Example

Assume that in a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the group.

Let the six people be $a_1, a_2, a_3, a_4, a_5, a_6$

$A = \{a_2, a_3, a_4, a_5, a_6\}$

$B = \{b_1, b_2\}$, where b_1 represents that one is a friend of a_1,...

By the generalized pigeonhole principle, of the five people in *A*, there are either three or more who are friends of a_1, or three or more who are enemies of a_1.

(1) Suppose that a_i, a_j, a_k are freinds of a_1

(2) Suppose that a_i, a_j, a_k are enemies of a_1

Ramsey theory (number)

Assume that in a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the group.

Solution: Let *A* be one of the six people. Of the five other people in the group, there are either three or more who are friends of *A*, or three or more who are enemies of *A*. This follows from the generalized pigeonhole principle, because when five objects are divided into two sets, one of the sets has at least $\lceil 5/2 \rceil = 3$ elements. In the former case, suppose that *B*, *C*, and *D* are friends of *A*. If any two of these three individuals are friends, then these two and *A* form a group of three mutual friends. Otherwise, *B*, *C*, and *D* form a set of three mutual enemies. The proof in the latter case, when there are three or more enemies of *A*, proceeds in a similar manner.

Ramsey theory (number)

- The Ramsey number $R(m,n)$, where m and n are positive integers greater than or equal to 2, denotes the minimum number of people at a party so that there are either m mutual friends or n mutual enemies, assuming that every pair of people at the party are friends or enemies.
- $R(3, 3) = 6$
- $R(m, n) = R(n,m)$
- $R(4, 4) = 18$
- $43 \leq R(5, 5) \leq 49$
- **Reinforced Generation of Combinatorial Structures: Ramsey Numbers**
- AlphaEvolve是Google DeepMind开发的进化式编码智能体（evolutionary coding agent）。其核心思想是”不直接求解问题，而是进化出求解问题的算法” AlphaEvolve的单一元算法同时刷新了5项经典Ramsey数下界，其中部分记录已保持长达20年之久。

Ramsey theory (number)

r \ s	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
3	6	9	14	18	23	28	36	40	47		61			82	92	100	106	111	122	132
4		18	25	36	49			92		128	139	148	159							
5					80	101	133													
6					115															

Table 1: Ramsey Number $R(r, s)$ lower bounds achieved by AlphaEvolve. Highlights indicate performance status: **light blue** for optimal lower bounds (matching upper bounds); **yellow** for matching current best-known lower bounds (where the upper bound remains strictly higher); and **green** for new results that surpass the prior best lower bound. Algorithms are hyperlinked.

Exercises

- **P. 426: 15, 24, 46, 47**

6.3 Permutations and Combinations

- The problems of permutation and combination: The coach has to select 5 players from 10 members of a tennis team, and prepare an ordered list of 4 players to play the four singles matches.

Permutations

- Permutation: A set of distinct objects in an *ordered arrangement* of these objects.
- *r*-permutation: An ordered arrangement of *r* elements of a set.
- The number of *r*-permutations of an *n*-element set is:

$$P(n,r) = n(n-1)(n-2)\ \ldots\ (n-r+1) = \frac{n!}{(n-r)!}$$

- Example: Consider the set $S = \{1,2,3\}$.
  - What are the 3-permutations of *S* ? 6
  - What are the 2-permutations of *S* ? 6

Examples

- How many different ways are there to select 4 different players from 10 players on a team to play four tennis matches, where the matches are ordered?

  $P(10,4)=10\times 9\times 8\times 7=5040$

- Suppose that there are eight runners in a race. Gold, silver and bronze medals are given. How many different ways are there to award these medals, if all possible outcomes of the race can occur?

  $P(8,3)=8\times 7\times 6=336$

- Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the sales woman use when visiting these cities?

  $7!=5040$

Combinations

- *r*-combination: An *unordered selection* of *r* elements of a set.
- The number of *r*-combinations of an *n*-element set, where $0 \le r \le n$, is:

$$C(n,r) = \frac{n!}{r!\,(n-r)!}$$

- Corollary: let *n* and *r* be nonnegative integers with $r \le n$. Then $C(n,r)=C(n,n-r)$

- Example: Consider the set $S=\{1,2,3\}$.
  - What are the 2-combinations of *S*? {1,2}, {1,3} ,{2,3}

Examples

- How many ways are there to select 5 players from a 10 member tennis team to make a trip to a match at another school?

  $C(10,5)=252$

- A committee has to be selected to develop a discrete mathematics course at a school.
  - The committee is to consist of 3 faculty members from the mathematics department and 4 from the computer science department.
  - There are 9 faculty members in the mathematics department and 11 in the computer science department.

  How many ways are there to select the committee?

  $C(9,3) \times C(11,4)=84 \times 330$

Summary

Order Relevant	Repetitions Allowed	Type of Result	Formula
Yes	Yes	Arrangement	n^r $n, r \geq 0$
Yes	No	Permutation	$P(n, r)=n!/(n-r)!$ $0 \leq r \leq n$
No	No	Combination	$C(n, r)=n!/[r!(n-r)!]$ $0 \leq r \leq n$

Combinatorial proof

- A combinatorial proof is a proof that uses counting arguments to prove a theorem, rather than some other method such as algebraic techniques.
- Example: $C(n,r)=C(n,n-r)$

  Proof: Suppose that S is a set with n elements. Every subset A of S with r elements corresponds to a subset of S with $n-r$ elements, namely A. Consequently, $C(n,r)=C(n,n-r)$.

Exercises

- **P.435: 25, 31, 35, 42, 44**

6.4 Binomial Coefficients

- This number represents the *binomial coefficient*:

$$\binom{n}{r}$$

- Because these numbers occur as coefficients in the expansion of powers of binomial expression such as $(a+b)^n$

Examples

$$(x+y)^0 = 1$$

$$(x+y)^1 = x+y$$

$$(x+y)^2 = x^2 + 2xy + y^2$$

$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

Binomial Theorem

- The binomial theorem gives the coefficients of the expansion of powers of binomial expressions.
- Binomial Theorem: Let *x* and *y* be variables, and let *n* be a positive integer.

$$(x+y)^n$$

$$= \sum_{j=0}^{n} C(n, j) x^{n-j} y^{j}$$

$$= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \binom{n}{2} x^{n-2} y^2 + \ldots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n$$

Examples

- Expand $(x+y)^4$.

  $X^4+4x^3y+6x^2y^2+4xy^3+y^4$

- What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?

  $C(25,13)=5,200,300$

- What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$?

  $C(25,12)2^{12}(-3)^{13}$

Corollary

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$$

$$\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n$$

Pascal's Triangle

- *Pascal's Triangle*: Geometric arrangement of binomial coefficients
- Pascal's Identity: Let n and k be positive integers with $n \geq k$. Then,

$$C(n+1, k) = C(n, k-1) + C(n, k)$$

- When two adjacent binomial coefficients in this Pascal's triangle are added, the binomial coefficient in the next row between these two coefficients is produced.
- Example: $C(7,5)=C(6,4)+C(6,5)$

Some other identities of the binomial coefficients

- Vandermonde's identity: let *m*, *n* and *r* be nonnegative integers with r not exceeding either *m* or *n*. Then

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}$$

Proof: Combinatorial proof

- Corollary: if n is a nonnegative integer, then

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2$$

Continue

$$\binom{n+1}{r+1} = \sum_{j=r}^{n} \binom{j}{r}$$

$$\sum_{k=r+1}^{n+1} \binom{k-1}{r} = \sum_{j=r}^{n} \binom{j}{r}$$

Proof: Use a combinatorial proof.

left: Counts the bit strings of length n+1

containing r+1 ones.

right: the possible locations of the final 1

in a string with r+1 ones

Exercises

- **P.443: 14, 26, 31, 36**

6.5 Generalized Permutations and Combinations

- How to solve counting problems where elements may be used more than once.
- How to solve counting problems in which some elements are indistinguishable.
- How to solve problems involving counting the ways to place distinguishable elements in boxes.
- Chapter 7 will discuss other counting problems.

Permutations with repetition

- Theorem: The number of r -Permutation of a set of n objects with repetition allowed is n^r.

- Example: How many strings of length n can be formed from the English alphabet ?

Combinations with repetition [1]

- Problem: How many ways are there to select four pieces of fruit from a bowl containing apples, oranges, and pears if the order in which the pieces are selected does not matter, only the type of fruit and not the individuals piece matters, and there are at least four pieces of each type of fruit in the bowl ?
- Problem: How many ways are there to select five bills from a cash box containing $1 bills, $2 bills, $5 bills, $10 bills, $20 bills, $50 bills, and $100 bills? Assume that the order in which the bills are chosen does not matter, that the bills of each denomination are indistinguishable, and that there are at least five bills of each type.

Combinations with repetition [2]

- Theorem: There are C(n+r-1,r) r-Combinations from a set with n elements when repetition of the element is allowed.
- Proof:

  1) Each r-Combinations from a set with n elements when repetition is allowed can be represented by a list of n-1 bars and r stars.

  2) The n-1 bars are used to mark off n different cells, with the ith cell containing a star for each time the ith element of the set occurs in the combination. For instance, a 6-combination of a set with four elements is represented with three bars and six stars. Here **|*||***

Combinations with repetition [3]

- Proof (cont.)

  3) Represents the combination containing exactly two of the first element,one of the second element and three of the fourth element of the set.from this model we get a combination containing n-1+r element, The number of choice is C(n-1+r,r), which corresponds to a choice of the r positions to place r stars from n-1+r positions.

Examples

- A cookie shop has 4 different kinds of cookies.how many different ways can 6 cookies be chosen?

  we can select many pieces from a kind of cookie, so the number of the ways to choose six cookies is the number of 6-combinations of a set with four elements. It is C(4-1+6,6)=C(9,3)

- How many solutions does the equation $x_1 + x_2 + x_3 = 11$ Have? Where x_1, x_2, and x_3 are nonnegative integers.

  It corresponds to a way to put 11 balls into 3 cells, so the answer is C(3-1+11,11)=C(13,2)=78

Examples

- How many solutions does the equation $x_1 + x_2 + x_3 = 11$ Have? Where $x_1 \geq 1$, $x_2 \geq 2$, and $x_3 \geq 3$.

  C(3+5-1,5)=21

- How many ways are there to place ten indistinguishable balls into eight distinguishable bins?

  The number of ways to place ten indistinguishable balls into eight bins equals the number of 10-combinations from a set with eight elements when repetition is allowed.

  C(8+10-1,10)=19448

Summary

Combinations and permutations with and without repetition

type	Repetition allowed	Formula
r-permutations	No	$\frac{n!}{(n-r)!}$
r-combinations	No	$\frac{n!}{r!(n-r)!}$
r-permutations	Yes	n^r
r-combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

Distributing Objects into Boxes

- **球盒问题 (balls and bins problem)**

球(n)	盒(m)	允许空	总放法数
可区分	可区分	是	
可区分	可区分	否	
可区分	不可区分	是	
可区分	不可区分	否	
不可区分	可区分	是	
不可区分	可区分	否	
不可区分	不可区分	是	
不可区分	不可区分	否	

Permutation with indistinguishable objects

- Problem: How many different strings can be made by reording the letters of the word SUCCESS ?

  C(7,3)C(4,2)C(2,1)C(1,1) = 7! / (3!2!1!1!)

- Theorem: The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, … , and n_k indistinguishable objects of type k , is

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Distinguishable Objects and Distinguishable Boxes

- Some counting problems can be solved by enumerating the ways distinguishable objects can be placed into distinguishable boxes.
- Problem: How many ways are there to distribute hands of 5 cards to each of four players from the standard deck of 52 cards ?

  C(52,5)C(47,5)C(42,5)C(37,5)=52! / (5!5!5!5!32!)
- Theorem: The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i, i=1,2,…,k, equals

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Indistinguishable Objects and Distinguishable Boxes

- Counting the number of ways of placing *n* indistinguishable objects into *k* distinguishable boxes turns out to be the same as counting the number of *n*-combinations for a set with *k* elements when repetitions are allowed.
- Example: How many ways are there to place 10 indistinguishable balls into eight distinguishable bins?

  C(8+10-1,10)

Distinguishable Objects and Indistinguishable Boxes

- Example: There are 14 ways to put four employees into three indistinguishable offices (*see Example* 10).
- There is no simple closed formula for the number of ways to distribute *n* distinguishable objects into *j* indistinguishable boxes.
- See the text for a formula involving *Stirling numbers of the second kind*.

Stirling numbers of the second kind

- Let $S(n, j)$ denote the number of ways to distribute n distinguishable objects into j indistinguishable boxes so that no box is empty. The numbers $S(n, j)$ are called **Stirling numbers of the second kind**.

$$\sum_{j=1}^{k} S(n, j)$$

$$S(n, j) = \frac{1}{j!} \sum_{i=0}^{j-1} (-1)^i \binom{j}{i} (j - i)^n$$

Stirling numbers of the second kind

- **Onto function**
- **Inclusion-Exclusion priniciple**
- **Inclusion-Exclusion and Special Numbers.pdf**
- **Stirling numbers.pdf**

Indistinguishable Objects and Indistinguishable Boxes

- Example: There are 9 ways to pack six copies of the same book into four identical boxes (*see Example* 11).
- The number of ways of distributing n indistinguishable objects into k indistinguishable boxes equals $p_k(n)$, the number of ways to write n as the sum of at most k positive integers in increasing order.
- No simple closed formula exists for this number.

Exercises

- **P.454: 15, 22, 31, 43, 48, 55, 59, 67**

6.6 Generating Permutations and Combinations

- Sometimes permutations or combinations need to be generated, not just counted.

Generating permutations

- Any set with n elements can be placed in one-to-one correspondence with the set {1,2,3,…,n}. We can list the permutations of any set of n elements by generating the permutations of the n smallest positive integers and then replacing these integers with the corresponding elements.
- Lexicographic ordering of the set of permutations of {1,2,3,…,n}: in this ordering, the permutation $a_1a_2\ldots a_n$ precedes the permutation of $b_1b_2\ldots b_n$, if for some k, with $1 \leq k \leq n$, $a_1=b_1$, $a_2=b_2$,…, $a_{k-1}=b_{k-1}$ and $a_k<b_k$.

Algorithm for generating the permutations

- An algorithm for generating the permutations can be based on a procedure that constructs the next permutation in lexicographic order following a given permutation .
  - Find the integer a_j and a_{j+1} with $a_j < a_{j+1}$ and $a_{j+1} > a_{j+2} > \ldots > a_n$.
  - The next largest permutation in lexicographic order is obtained by putting in the jth position the least integer among $a_{j+1}, a_{j+2}, \ldots, a_n$ that is greater than a_j and listing in increasing order the rest of the integers $a_j, a_{j+1}, a_{j+2}, \ldots, a_n$ in positions $j+1$ to n.

Examples

- Generate the permutations of the 1,2,3,4 in lexicographic order

$$1\ 2\ 3\ 4 \rightarrow 1\ 2\ 4\ 3 \rightarrow 1\ 3\ 2\ 4 \rightarrow 1\ 3\ 4\ 2 \rightarrow 1\ 4\ 2\ 3 \rightarrow 1\ 4\ 3\ 2$$
$$\rightarrow 2\ 1\ 3\ 4 \rightarrow 2\ 1\ 4\ 3 \rightarrow 2\ 3\ 1\ 4 \rightarrow 2\ 3\ 4\ 1 \rightarrow 2\ 4\ 1\ 3 \rightarrow 2\ 4$$
$$3\ 1 \rightarrow 3\ 1\ 2\ 4 \rightarrow 3\ 1\ 4\ 2 \rightarrow 3\ 2\ 1\ 4 \rightarrow 3\ 2\ 4\ 1 \rightarrow 3\ 4\ 1\ 2 \rightarrow$$
$$3\ 4\ 2\ 1 \rightarrow 4\ 1\ 2\ 3 \rightarrow 4\ 1\ 3\ 2 \rightarrow 4\ 2\ 1\ 3 \rightarrow 4\ 2\ 3\ 1 \rightarrow 4\ 3\ 1$$
$$2 \rightarrow 4\ 3\ 2\ 1$$

- What is the next largest permutation in lexicographic order after 362541 ?

364125

Generating all combinations of the elements of a finite set

- Correspondence between subsets of $\{a_1,a_2,\ldots,a_n\}$ and bit strings of length n. A bit string of length n is also the binary expansion of an integer between 0 and 2^n-1. At each stage the next largest expansion is found by locating the first position from the right that is not a 1, then changing all the 1s to the right of this position to 0s and making this first 0 as 1.
- Example: Find the next largest bit string after 1000100111 ?

1000101000

Algorithm for generating the *r*-combinations of the set {1,2,3,…,*n*}

- An *r*-combinations can be represented by a sequence containing the elements in the subset in increasing order. The *r*-combinations can be listed using lexicographic order on these sequences.
- The next combinations after $a_1 a_2 \ldots a_r$ can be obtained by:
  - **First, locate the last element a_i in the sequence such that $a_i \neq n - r + i$.**
  - **Then, replace a_i with $a_i + 1$ and a_j with $a_i + j - i + 1$, for $j = i+1, i+2, \ldots, r$.**

Example

- Find the next largest 4-combination of the set {1,2,3,4,5,6} after {1,2,5,6}

  The last term among the terms a_i with a_1=1, a_2=2, a_3=5, and a_4=6 such that $a_i \neq 6\text{-}4+i$ is a_2=2.

  hence the next 4-combination is {1,3,4,5}

$a_i \neq n - r + i$.
Then, replace a_i with a_i +1 and a_j with $a_i + j - i+1$, for $j = i+1, i+2,\ldots,r$.

Exercises

- **P.461: 5, 8, 9**